

University of Asia Pacific

Department of Computer Science and Engineering

CSE 402:Operating Systems Lab

LAB REPORT

Report Number:05

Title: SSTF Disk Scheduling

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1. Problem Statement

Write a Python program to implement the Shortest Seek Time First (SSTF) disk scheduling algorithm. The program should take the initial disk head position and a sequence of disk I/O requests as input. It should select the pending request closest to the current head position and serve

it first until all requests have been completed. Finally, calculate and display the seek sequence and

total head movement of the disk head.

head = 50

requests = [60, 20, 100, 10, 15, 22, 42]

2. Source Code

```python
def sstf(requests, head):
    requests = requests.copy()
    total_seek = 0
    seek_sequence = []
    current_head = head

    while requests:
        closest = min(requests, key=lambda x: abs(x - current_head))

        total_seek += abs(closest - current_head)
        current_head = closest

        seek_sequence.append(closest)
        requests.remove(closest)

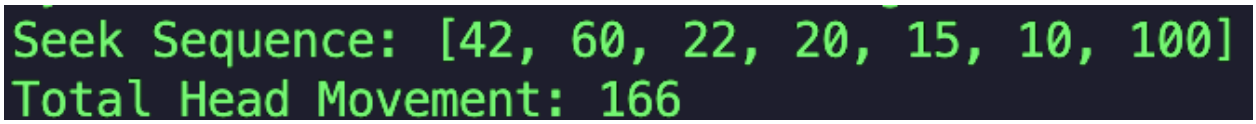
    print("Seek Sequence:", seek_sequence)
    print("Total Head Movement:", total_seek)

head = 50

requests = [60, 20, 100, 10, 15, 22, 42]

sstf(requests, head)
```

3. Output ScreenShot



4. Discussion

In this lab, the SSTF disk scheduling algorithm was implemented using Python. A predefined initial head position and a list of disk requests were used as input. The program selected the request closest to the current head position, calculated the seek distance, and repeated the process until all requests were served. Finally, the seek sequence and total head movement were displayed.

5. Conclusion

Through this lab, the working principle of the SSTF disk scheduling algorithm was understood and implemented successfully. The experiment demonstrated how selecting the nearest disk request can reduce unnecessary head movement. However, the algorithm may cause starvation for requests located far from the current head position.

GitHub Repo link: https://github.com/MasfiRahman/CSE-402-Operating-System-Lab-.git